The two types of spores are microspores and megaspores. **Microspores** grow into a male gametophyte, and **megaspores** grow into a female gametophyte (see Fig. 23.18).

Check Your Progress 23.3

1. Name two features of lycophytes that increase their ability to survive on land.
2. Explain how xylem contributes to an upright body plant.
3. Define the terms *homosporous* and *heterosporous*.

23.4 Evolution of Pteridophytes: Megaphylls

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify three types of pteridophytes.
2. Compare and contrast microphylls and megaphylls.
3. Identify the components of the ferns' life cycle.

Pteridophytes is a broad term used to describe a group of seedless vascular plants that includes ferns and their allies, the horsetails and whisk ferns. Both pteridophytes and the seed plants have megaphylls. **Megaphylls** are broad leaves with several strands of vascular tissue. Figure 23.12*a* shows the difference between microphylls and megaphylls, and Figure 23.12*b* shows how megaphylls could have evolved.

Megaphylls, which evolved independently from the evolution of microphylls, allow plants to efficiently collect solar energy, leading to the production of more food and the possibility of producing more offspring than plants without megaphylls. Therefore, the evolution of megaphylls made plants more fit. Recall that fitness, in an evolutionary sense, is judged by the number of living offspring an organism produces relative to others of its own kind.

The pteridophytes, like the lycophytes, were dominant from the late Devonian period through the Carboniferous period. Today, the lycophytes are quite small, but some of the extinct relatives of today's club mosses were 35 m tall and dominated the Carboniferous swamps. The horsetails (around 18 m tall) and ancient tree ferns (8 m or more in height) also contributed significantly to the great swamp forests of the time (see the following Evolution feature, "Carboniferous Forests").

Horsetails

Today, **horsetails** consist of one genus, *Equisetum,* and approximately 15 species of distinct seedless vascular plants. Most horsetails inhabit wet, marshy environments around the globe. About 299 MYA, horsetails were dominant plants and grew as large as

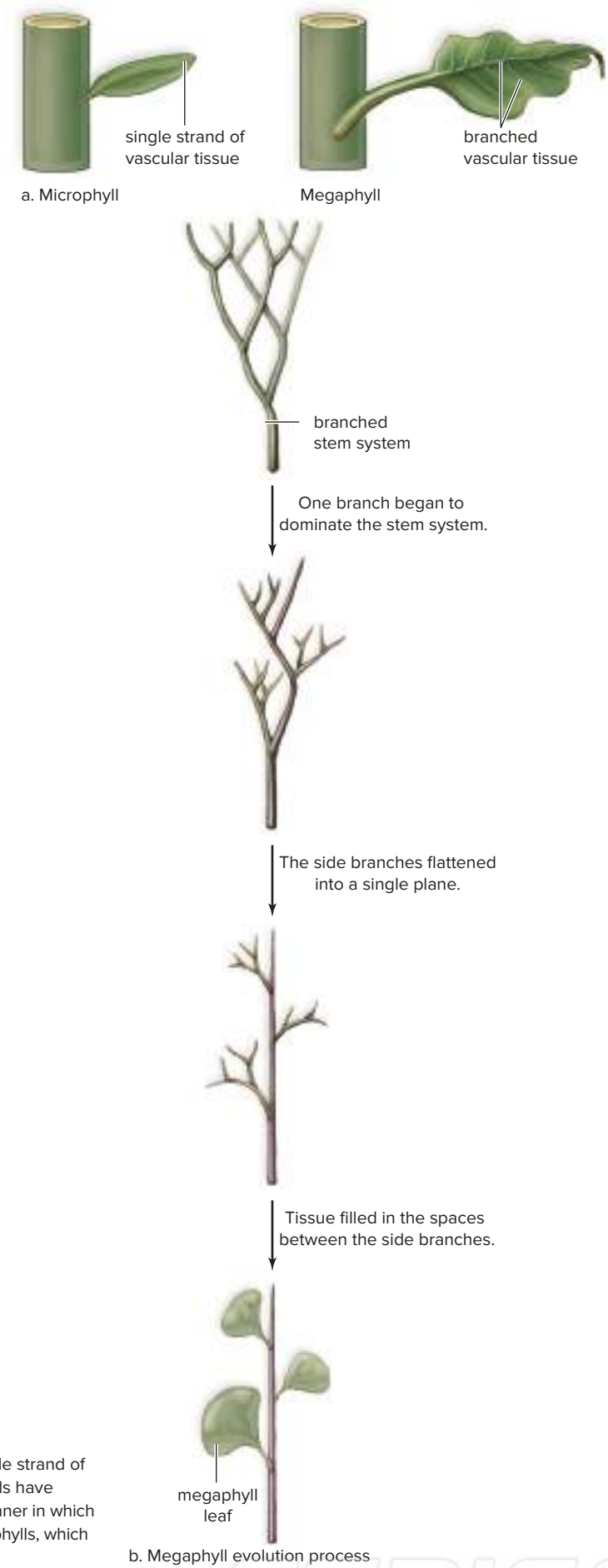


Figure 23.12 Microphylls and megaphylls. **a.** Microphylls have a single strand of vascular tissue, which explains why they are quite narrow. In contrast, megaphylls have several branches of vascular tissue and are broader. **b.** These steps show the manner in which megaphylls may have evolved. All vascular plants, except lycophytes, bear megaphylls, which can gather more sunlight and produce more organic food than microphylls.

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THEME Evolution

Carboniferous Forests

Our industrial society runs on fossil fuels, such as coal. The term *fossil fuel* might seem odd at first until one realizes that it refers to the remains of organic material from ancient times. During the Carboniferous period, around 299 million years ago, a great swamp forest (Fig. 23A) encompassed what is now northern Europe, the Ukraine, and the Appalachian Mountains in the United States. The weather was warm and humid, and the plants grew very tall. These were not plants that would be familiar to us. Instead, they were related to various groups of plants that are less well-known—the lycophytes, horsetails, and ferns.

Lycophytes today may stand as high as 30 cm, but their ancient relatives were 35 m tall and 1 m wide. The strobili were up to 30 cm long, and some had leaves more than 1 m long. Horsetails, too—at 18 m tall—were giants compared to today's specimens. Tree ferns were also taller than the tree ferns found in the tropics today. The progymnosperms, including "seed ferns," so named because their leaves looked like fronds, were significant plants of a Carboniferous swamp.

The amount of biomass in a Carboniferous swamp forest was enormous, and occasionally the swampy water rose, covering the plants that had died. Dead plant material under water does not decompose well. The partially decayed remains became covered by sediment, which changed them into sedimentary rock. Exposed to significant amounts of pressure and over long periods of time, the organic material became coal. This process continued for millions of years, resulting in immense deposits of coal. Geological upheavals raised the deposits to the levels where they can be mined today.

With a change of climate, many of the plants of the Carboniferous period became extinct. Some of their smaller herbaceous relatives have evolved and survived to our time. We owe the industrialization of today's society to these ancient forests.

Questions to Consider

1. How might a geologist use this information to look for new sources of coal?
2. Why is coal not considered to be a renewable resource?



a.



b.

Figure 23A Swamp forest of the Carboniferous period. **a.** Nonvascular plants, early vascular plants, and early gymnosperms (often called "progymnosperms") dominated the swamp forests of the Carboniferous period. **b.** Micrograph of an early gymnosperm called a seed fern. (a): Richard Bizley/Science Source; (b): Sinclair Stammers/Science Source

modern trees. Today, horsetails have a rhizome that produces hollow, ribbed aerial stems and reach a height of 1.3 m (Fig. 23.13).

Equisetum can be branched or unbranched, with leaves that may have been megaphylls at one time but now are reduced and form whorls at the nodes. The skirtlike, slender, green side branches make the plant bear a resemblance to a horse's tail. Many horsetails have strobili at the tips of all stems; others send up buff-colored stems that bear the strobili. The spores germinate into inconspicuous and independent gametophytes. *Equisetum* spores are sensitive to humidity and are "spring-loaded." When conditions are right, the spores have been known to be ejected and travel upward of 2 miles in an hour.

The stems are tough and rigid because of silica deposited in cell walls. Early Americans, in particular, used horsetails for scouring pots and called them "scouring rushes." Today, they are still used as ingredients in a few abrasive powders.

Whisk Ferns

Whisk ferns are represented by the genera *Psilotum* and *Tmesipteris,* which are native to tropical and subtropical regions. The two *Psilotum* species resemble a whisk broom (Fig. 23.14), because they have no leaves. A horizontal rhizome gives rise to aboveground stems that repeatedly fork. The pumpkin-shaped sporangia are borne

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